

1.
CO1

- a) A processor operates at a clock rate of 3 GHz and achieves a performance of 1500 MIPS while executing a program with an instruction count of 900×10^6 . The instruction mix and individual CPI values are given below:

Instruction Type	Fraction	CPI
R-Type	50%	1
<u>Load</u>	30%	3
Store	10%	5
Jump	10%	?

Answer the following questions based on the given information:

- Determine the CPI of the Jump instruction.
- Calculate the execution time of the program in seconds.
- A handy tool revealed that Load instructions account for 40% of the execution time. A hardware engineer claims that "redesigning the cache to reduce the load CPI by 3 times will reduce the total execution time by at least 45%." Verify with calculations whether this statement is true or false.

- b) A data center manager is reviewing infrastructure costs and notices that their storage system uses three times more disk space than their actual requirement, with the same data being copied into multiple different locations. The vendor justifies this as "an architectural necessity, not inefficiency." The manager is not convinced and proposes switching to a cheaper single-copy storage system to cut costs.

Explain briefly why the manager's proposal is risky and which architectural idea the vendor is referring to.

2.
CO1

- a) For the following questions, consider 6 digits after the floating point while converting from decimal to binary.

- Consider $X = 6.2571 \times 10^1$ and $Y = 0.17562 \times 10^2$. Perform $X+Y$ using floating point representation. You must show all the necessary steps, and the final answer must be in decimal.
- Let's assume we will store the final answer in IEEE754 24-bit format, where the fraction size is 17-bit. Mention if there is any overflow or underflow in the answer.

- b) Determine whether the following statements are true or false, if the statement is false, then write the correct answer.

- The smallest number represented in a 10-bit IEEE754 Floating Point Representation system with 6-bit fraction is -1.11111×2^3
- Rounding can change the sign of a number.

3. CO2 a) Consider the following partially incomplete RISC-V program. Two instructions are missing at addresses 60040 and 60056. Their corresponding machine codes are provided.

Address(Decimal)	Instruction	Machine code
60032	addi x6, x0, 1	
60036	and x9, x18, x6	
60040		0x00649A4D
60044	lh x5, 0(x16)	
60048	addi x5, x5, -1	
60052	sh x5, 0(x16)	
60056		0x0000284D
60060	lh x5, 2(x16)	
60064	srl x5, x5, 1	
60068	sh x5, 2(x16)	
60072		

3
+
2
+
1

Assume that the missing instructions at addresses 60040 and 60056 are **branch** instructions. Opcode for any branch instruction = 1001101. Funct3 for beq = 010, bne = 001, bge = 000, blt = 011.

Answer the following questions based on the given information above:

- i) Decode the machine codes stored at addresses 60040 and 60056 into their corresponding assembly instructions.

Note: Use "label1" as the label for the first branch instruction and "label2" as the label for the second branch instruction. Write your final answers in the table above.

- ii) Determine the target address of each branch instruction by showing the necessary calculations.
iii) Suppose the target address of the second branch instruction was 67212 instead of the one obtained in part (ii). Would branching to that address be possible? Justify your answer.

- b) A student wants to add the 16-bit value stored at memory location (x10) with register x6 and store the result in x5. They wrote the following instruction:

add x5, 0(x10), x6

Briefly explain if this is a valid instruction. If not, write the correct instruction.

- c) The high-level code below computes the sum of a word array of length n. The function uses registers x8 and x9 for local variables i and sum, respectively. The base address of the array (arr[]) is saved in the x10 register, and the argument n is in x15. Your friend wrote the RISC-V assembly translation for this C code. However, it contains 11 errors.

Find each of the faulty instructions and write the corrected version in the table below. The first one has been done for your reference.

1
+
1

5

High-level Code	Faulty RISC-V assembly
<pre> int array_sum(int arr[], int n) { int i = 0, sum = 0; while (i < n) { sum += arr[i]; i++; } return sum; } </pre>	<pre> array_sum: addi x2, x2, 16 sw x8, 8(x2) sw x9, 0(x2) addi x8, x0, 0 add x9, x0, 0 loop: bge x15, x8, loop_exit srli x5, x8, 2 add x5, x10, x5 lw x6, 0(x5) add x9, x9, x6 addi x8, x8, -1 bne x0, x0, loop loop_exit: add x12, x9, x0 ld x9, 0(x2) ld x8, 8(x2) addi x2, x2, -8 jalr x1, 0(x0) </pre>

Error no.	Faulty instruction	Corrected Instruction
0	addi x2, x2, 16	addi x2, x2, -16
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

d) Write a RISC-V assembly program that divides a number stored in register x19 by the number stored in register x20 and stores:

- the quotient in register x15
- the remainder in register x19

Note: You are not allowed to use the div instruction, and you may assume that the divisor is non-zero